Business Programming – Project Guide

General Information

The project must be developed in groups of **4 students**, preferably. Groups of **3 students** may be accepted when necessary.

Each group must choose **one** of the following options:

- **Option A — Event Registration Web App**
- **Option B — Low-Code / No-Code Business App**
- **Option C — Competitor Price Automation**
- **Option D — Project Management and Version Control**

The project should demonstrate how programming, software tools, automation, or project management can be applied to a realistic business problem.

Option A — Event Registration Web App

Create a small web application for an event, such as a fair, conference, workshop, seminar, or networking event.

People must be able to register for the event, and the organization must be able to manage the registrations.

Minimum Functionality

The application must include:

- Event information page;
- Registration form;
- Database to store registrations;
- List of registered participants;
- Detail page for each registration;
- Edit or cancel a registration;

- Basic HTML/CSS styling;
- At least one simple JavaScript interaction.

The database should include at least **two related tables**, for example:

- Event;
- Registration.

Technologies

Python, Flask, SQLite, HTML, CSS, JavaScript.

Option B — Low-Code / No-Code Business App

The same as option A, but with LowCode /NoCode (shown in week 10)

The deliverables are the same as option A

Option C — Competitor Price Automation

Create a Python automation that collects and compares competitor prices. Choose a business area for this and products of that area.

The objective is to help a business understand prices in the market and support pricing decisions.

Minimum Functionality

The project must include:

- At least **20 products**;
- At least **3 different online shops**;
- Product name, shop, URL, price, and collection date;
- Automated price collection using Python;
- Data storage in CSV, Excel, JSON, or SQLite;
- Price comparison between shops;

- Identification of lowest price, highest price, and price difference;
- Final business report with tables and at least one chart.

You must collect data responsibly: Do not collect personal data; Do not overload websites (use pause); Mention the websites used as sources.

If scraping is not possible, you may use an API, public data, or another approved alternative.

Option D — Project Management and Version Control

Plan a realistic software project for a business and create a small technical prototype.

This option focuses on software development methodology, collaboration, GitHub, and project organization.

Minimum Requirements

The project must include:

1. Project Management

Explain:

- The business problem;
- The proposed software solution;
- Main requirements;
- Waterfall and Agile/Scrum;
- Which methodology is more appropriate and why;
- A simple backlog;
- A sprint plan;
- A Trello, GitHub Projects, Jira, or similar board.

Use citations from credible sources.

2. Version Control

Create a GitHub repository with:

- At least **5 meaningful commits**;
- A README file;
- At least **4 project files**;
- Clear commit messages.

3. Small Prototype

Create a small prototype related to the planned project, for example:

- A very small Flask app;
- A Python script;
- A simple website;
- A low-code prototype;
- A simple mobile app prototype.

Deliverables for All Options

Each group must submit in Moodle, a zip file contaning:

1. Code / data folder

The code/ data folder should include:

- Source code or exported project files;
- Data files, if applicable;
- Output files, if applicable;
- README file;
- Any files needed to understand or run the project.

2. Short Report

Maximum **8 pages**, excluding appendices.

The report should include:

- Project title;
- Group members;
- Selected option;
- Business problem;
- Solution developed;
- Main features;
- Technologies or tools used;
- Screenshots;
- Testing evidence;
- Limitations and future improvements;
- AI-use statement, if AI tools were used.

3. Video Demonstration

Maximum **7 minutes**.

The video should show:

- The problem;
- The solution working;
- Main features;
- Technical explanation;
- What each student contributed.

4. Individual Contribution

You must include the **excel** with the groups contribution filled:

AI-Use Statement

The use of AI tools is allowed, but it must be explained.

The group must state in the report:

- Which AI tools were used;
- For what tasks;
- What was accepted;
- What was corrected;
- What the group learned.

Example:

“We used ChatGPT to help generate an initial version of a Flask route. The code was tested and corrected by the group before being included in the final project.”

Advice

A simple, complete, and well-explained project is better than an ambitious project that does not work or cannot be explained by the group.